D Chemistry

Fundamental Concepts

Q.1 When one mole of gaseous hydrogen ions are dissolved in water to form an infinitely dilute solution, the amount of heat liberated is

A) -1891 $kJmol^{-1}$
B) -1075 $kJmol^{-1}$✓
C) -499 $kJmol^{-1}$
D) -1562 $kJmol^{-1}$

Q.2 One mole of any gas at standard temperature and pressure (STP) occupies a volume of

A) 20.414 dm^3
B) 22.414 dm^3✓
C) 22.414 cm^3
D) 23.414 dm^3

Q.3 The relative abundance of the isotopes of the elements can be determined by:

A) Mass Spectrometry✓
B) X-rays
C) Chromatography
D) Solvent Extraction

Q.4 The root mean square velocity of gases is inversely proportional to the square root of their:

A) Molar mass✓
B) Temperature
C) Pressure
D) Volume

Q.5 The acid which can be purified by the sublimation is:

A) Acetic Acid
B) Benzoic Acid✓
C) Oxalic Acid
D) Citric Acid

Q.6 Paper chromatography is used for:

A) Elemental Analysis
B) Industrial Purification
C) Qualitative Analysis✓
D) Structural Analysis

Q.7 The process of filtration is used to separate ________ particles from liquids.

A) Radial.
B) Angular.
C) Insoluble.✓
D) Soluble.

Q.8 London forces are very significant in ________

A) Sulphur.
B) Phosphorous.
C) Argon.✓
D) Sugar.

Q.9 Ten moles of hydrogen are allowed to react with 6 moles of oxygen. How much water will be obtained from reaction on complete consumption of one gas?

A) 10 moles.✓
B) 8 moles.
C) 6 moles.
D) 4 moles.

Q.10 In mass spectrometer, detector or collector measures the:

A) Masses of isotopes
B) Percentages of isotopes
C) Relative abundances of isotopes✓
D) Mass numbers of isotopes

Q.11 How many 'Cl' (chlorine) atoms are in two moles of chlorine?

A) $2 \times 6.02 \times 10^{-23}$ atoms
B) $35.5 \times 6.02 \times 10^{23}$ atoms
C) 2×10^{23} atoms
D) $2 \times 6.02 \times 10^{23}$ atoms✓

Q.12 An organic compound has empirical formula C_3H_3O, if molar mass of compound 110.15 $gmol^{-1}$. The molecular formula of this organic compound is (A, of C=12, H=1.0 and O=16)

A) $C_6H_6O_2$✓
C) C_3H_3O
C) $C_9H_9O_3$
D) $C_6H_6O_3$

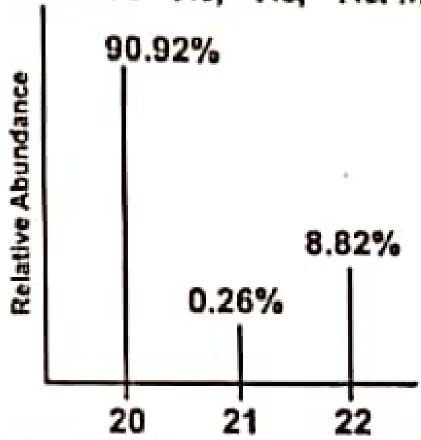
Q.13 When 8 grams (4 moles) of H_2 react with 2 moles of O_2, how many moles of water will be formed?
A) Five
B) Four✓
C) Six
D) Three

Q.14 Hydrogen burns in chlorine to produce hydrogen chloride. The ratio of masses of reactants in chemical reaction is:

$$H_2 + Cl_2 \longrightarrow 2HCl$$

A) 1:35.5✓
B) 2:35.5
C) 1:71
D) 2:70

Q.15 A sample of Neon is found to exist as ^{20}Ne, ^{21}Ne, ^{22}Ne. Mass spectrum of 'Ne' is as follow:



What is the relative atomic mass (A_r value) of Neon?
A) 20.18
B) 20.10
C) 20.28✓
D) 20.22

Q.16 A polymer of empirical formula CH2 has molar mass of 28000 g mol^{-1}. Its molecular formula will be
A) 100 times that of its empirical formula
B) 200 times that of its empirical formula
C) 500 times that of its empirical formula
D) 2000 times that of its empirical formula✓

Q.17 The number of molecules in 9 g of ice (H_2O) is
A) 6.02×10^{24}
B) 6.02×10^{23}
C) 3.01×10^{24}
D) 3.01×10^{23}✓

Q.18 How many moles of sodium are present in 0.1 g of sodium?
A) 4.3×10^{-3}✓
B) 4.03×10^{-1}
C) 4.01×10^{-2}
D) 4.3×10^{-2}

Q.19 With the help of spectral data given calculate the mass of Neon and encircle the best option. (Percentage of $10Ne^{20}$, $10Ne^{21}$ and $10Ne^{22}$ are 90.92%, 0.26% and 8.82% respectively).
A) 22.18 amu
B) 21.18 amu
C) 20.18 amu✓
D) 22.20 amu

Q.20 The substance for the separation of isotopes is firstly converted into the:
A) Neutral state
B) Vapour state✓
C) Free state
D) Charged state

Q.21 The number of moles of CO2 which contain 8.00 gm of oxygen is:
A) 0.75
B) 0.25✓
C) 1.50
D) 1.00

Q.22 There are almost 200 million people alive in Pakistan. If you were to distribe Rupee 100 to each Pakistani in the form of 5 Rupee coin, how many moles of coins you must have:
A) 6.67×10^{-14}
B) 1.15×10^{-14}
C) 6.67 x 1014
D) 1.5×10^{14}

States of Matter

Q.1 Collagen and albumin are
A) Simple proteins✓
B) Derived proteins
C) Polyamides
D) Polysaccharides

Q.2 If we are given the mass of one substance, we can calculate volume of other substances and vice a versa with the help of balanced chemical equation. This is called
A) Mass-mass relationship
B) Mass-mole relationship
C) Mole-volume relationship
D) Mass-volume relationship✓

Q.3 In the process of respiration there is application of:
A) Dalton's Law✓
C) Charles's Law
C) Boyle's Law
D) Graham's Law

Q.4 ________% of the known universe is in the plasma state.
A) 30
B) 99✓
C) 50
D) 80

Q.5 100 m^3 of a gas at 3 atm pressure and 27 °C is transferred to a chamber of 300 m^3 volume maintained at a temperature of 327 °C. What will be the pressure in chamber?
A) 6 atm.
B) 4 atm.
C) 2 atm.✓
D) 1 atm.

Q.6 Melting point of water is higher than petrol, because intermolecular forces in water are:
A) Weaker than petrol
B) Stronger than petrol✓
C) Same as in petrol
D) Negligible

Q.7 DNA molecule is double stranded, in which two chains of DNA are twisted around each other by:
A) Hydrogen bonds✓
B) Vander Waal's force
C) Covalent bonds
D) Dative bonds

Q.8 The number of molecules in 22.4 dm^3 of H2 gas at 0 °C and 1 atm are
A) 60.2×10^{23}
B) 6.02×10^{22}
C) 6.02×10^{25}
D) 6.02×10^{23}✓

Q.9 Correct order of boiling points of the given liquid is
A) $H_2O > HF > HCl > NH_3$
B) $HF > H_2O > HCl > NH_3$
C) $H_2O > HF > NH_3 > HCl$✓
D) $HF > H_2O > NH_3 > HCl$

Q.10 The coordination number of Na^+ in NaCl crystal is:
A) 6✓
B) 2
C) 4
D) 8

Q.11 There are four gases H_2, He, N_2 and CO_2 at 0 °C. Which gas shows greater non-ideal behavior?
A) He
B) CO_2✓
C) H_2
D) N_2

Q.12 Ice is less dense than water at:
A) 0 °C✓
B) 4 °C
C) -4 °C
D) 2 °C

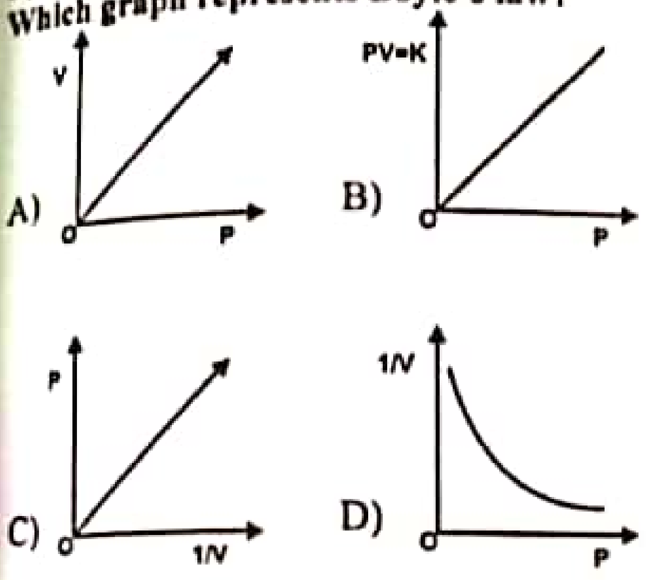
Q.13 At a given temperature and pressure, the one which shows marked deviation from ideal behavior is
A) N_2
B) N_3
C) CO_2✓
D) He

Q.14 If the volume of a gas collected at a temperature of 600 °C and pressure of 1.05×10^5 Nm^{-2}

³ is 60 dm³, what would be the volume of gas at STP ($P = 1.01 \times 10^5\ Nm^{-2}$, $T = 273$ K)?

A) 25 cm^3
C) 100 cm^3
B) 75 cm^3
D) 51 cm^3 ✓

Q.15 Which graph represents Boyle's law?



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Q.16 London dispersion forces are the only forces present among the:

A) Molecules of H_2O in liquid state
B) Molecules of HCl gas
C) Atoms of helium in gaseous state at high temperature ✓
D) Molecules of solid chlorine

Atomic Structure

Q.1 In an electrochemical series, standard electrode potentials are arranged on the basis of:

A) pH scale
C) Hydrogen Scale ✓
B) pOH scale
D) pK_a scale

Q.2 Urea is produced by the reaction of liquid ammonia with

A) CO_2 ✓
C) CaO
B) CO
D) C

Q.3 Which one of the following mathematical expressions represents the Avogadro's law?

A) $V = R\dfrac{nT}{P}$ (when 'T' and 'n' are constant)

B) $V = R\dfrac{nT}{P}$ (when 'P', 'T' and 'n' are constant)

C) $V = R\dfrac{P}{nT}$ (when 'P' and 'n' are constant

D) $V = R\dfrac{nT}{P}$ (when 'P' and 'T' are constant) ✓

Q.4 The charge of one gram of electron is

A) 1.7588×10^{-11}
C) 1.602×10^{-19}
B) 1.7588×10^{11}
D) 1.7588×10^{8} ✓

Q.5 Which quantum number helps to study the orientation of an orbital in space?

A) Principal Quantum Number
B) Spin Quantum Number
C) Magnetic Quantum Number✓
D) Azimuthal Quantum Number

Q.6 Metallic conduction involves the relatively free movement of their ______ through the metallic lattice
A) Atoms
B) Molecules
C) Electrons✓
D) Ions

Q.7 . During isotopic analysis, the pressure of the vapours of the ions maintained in ionization chamber of mass spectrometer is:
A) Around 10^{-7} torr✓
B) Around 10^{-3} torr
C) 1 torr
D) 10^{-7} torr

Q.8 Electron gas theory was proposed to explain the bonding in ______ solids:
A) Molecular
B) Ionic
C) Covalent
D) Metallic✓

Q.9 The electron present in a particular orbit ______ energy.
A) Releases.
B) Does not radiate.✓
C) Absorbs.
D) None of these.

Q.10 The elements for which the value of ionization energy is low, can:
A) Gain electrons readily
B) Gains electron with difficulty
C) Loss electrons less readily
D) Lose electrons readily✓

Q.11 The nature of cathode rays in discharge tube:
A) Depends on the nature of gas taken in the discharge tube
B) Depends upon the nature of cathode in discharge tube
C) Is independent of the nature pf the gas in discharge tube✓
D) Depends upon the nature of anode in the discharge tube

Q.12 The relative energies of 4s, 4p and 3d orbitals are in the order
A) 3d < 4p <4s
B) 4s < 3d < 4p✓
C) 4p < 4s < 3d
D) 4p < 3d < 4s

Q.13 With increase in the value of Principal Quantum Number 'n', the shape of the s-ort remains the same although their sizes
A) Decrease
B) Increase✓
C) Remain the same
D) May or may not remain the same

Q.14 Correct order of energy in the given subshells is:
A) 5s > 3d > 3p > 4s
B) 5s > 3d > 4s > 3p✓
C) 3p > 3d > 5s > 4s
D) 3p > 3d > 4s > 5s

Q.15 Number of electrons in the outermost shell of chloride ion (Cl^-) is:
A) 17
B) 3
C) 1
D) 8✓

Q.16 According to the number of protons, neutrons and electrons given in the table, whic of the following options is correct?

Species	Proton	Neutron	Electron
As	33	42	30
Ga	31	39	28
Ca	20	20	20

A) As^{-3}, Ga^{+3}, Ca
B) As^{+1}, Ga^{+2}, Ca
C) As^{+3}, Ga^{+3}, Ca^{+2}✓
D) As^{+1}, Ga, Ca^{+2}

Q.17 If the e/m value of electron is 1.7588×10^{11} coulombs Kg^{-1}, then what would be the ma electron in grams (charge on electron is 1.6022×10^{-19} coulombs)?

A) 9.1095×10^{-31} g✓
B) 91.095×10^{-31} g
C) 9.1095×10^{-28} g
D) 0.919095×10^{-31} g

Q.18 Which one of the following pairs has the same electronic configuration as possessed by Neon (Ne-10)?
A) Na^+, Cl^-
B) K^+, Cl^-
C) Na^+, Mg^{2+}
D) Na^+, F^- ✓

Q.19 There are four orbitals s, p, d and f. Which order is correct with respect to the increasing energy of the orbitals?
A) 4s < 4p < 4d < 4f✓
B) 4p < 4s < 4f < 4d
C) 4s < 4f < 4p < 4d
D) 4f < 4s < 4d < 4p

Q.20 Number of neutrons in $^{66}_{30}Zn$ will be:
A) 30
B) 35
C) 38
D) 36✓

Chemical Bonding

Q.1 Energy required to remove an electron from the outermost shell of its isolated gaseous atom in the ground state is
A) Electron affinity
B) Lattice energy
C) Ionization energy✓
D) Crystal energy

Q.2 Paramagnetic behavior of an atom, ion or molecule is due to presence of
A) Unpaired electrons✓
B) Paired electrons
C) Protons
D) Neutrons

Q.3 In the structure of NaCl, each Na^+ is surrounded by _____ Cl^- ions.
A) Four
B) Eight
C) Five
D) Six✓

Q.4 The ionization energy of hydrogen atom is
A) Zero
B) 13.13 $kJmol^{-1}$
C) 1313.31 $kJmol^{-1}$✓
D) 1313.31 k^2Jmol

Q.5 The inter-ionic distance in a crystal lattice of KCl is
A) 314 pm✓
B) 181 pm
C) 95 pm
D) 300 pm

Q.6 The number of bonds in nitrogen molecule is
A) One σ and two π✓
C) One σ and one π
C) Three σ only
D) Two σ and one π

Q.7 The extent of bonding of a light ray after passing through prism depends upon:
A) Wavelength of photons✓
B) Wave number of photons
C) Energy of photons
D) Frequency of photons

Q.8 Splitting of spectral lines in closely spaced lines in presence of magnetic field is called:
A) Stark Effect
B) Zeeman Effect✓
C) Photoelectric Effect
D) Compton Effect

Q.9 A bond is not formed:
A) When both forces become equal to each other
B) When repulsive forces become equal to zero
C) When attraction forces dominate repulsive forces
D) When repulsive forces dominate attraction forces✓

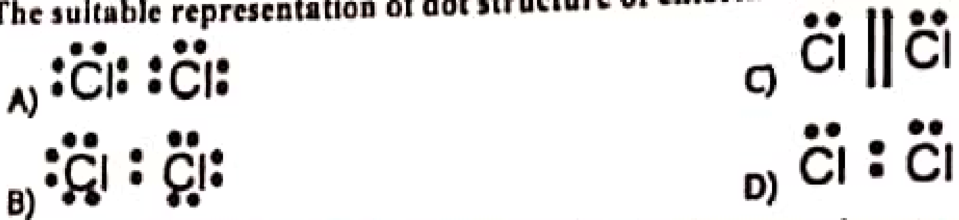
Q.10 If the electronegativity difference between bonded atoms is zero, the bond between the two atoms is:

A) Polar
B) Partially Ionic
C) Non-polar✓
D) Both B and C

Q.12 In crystal lattice of ice, each O-atom of water molecule is attached to:

A) Four H-atoms✓
B) Three H-atoms
C) One H-atom
D) Two H-atoms

Q.13 The suitable representation of dot structure of chlorine molecule is:



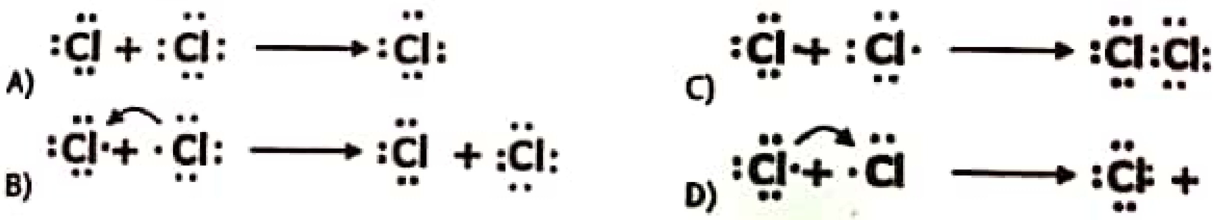
Q.14 When the two partially filled atomic orbitals overlap in such a way that the probat finding electron is maximum around the line joining the two nuclei, the result formation of

A) Sigma Bond✓
B) Pi-Bond
C) Hydrogen Bond
D) Metallic Bond

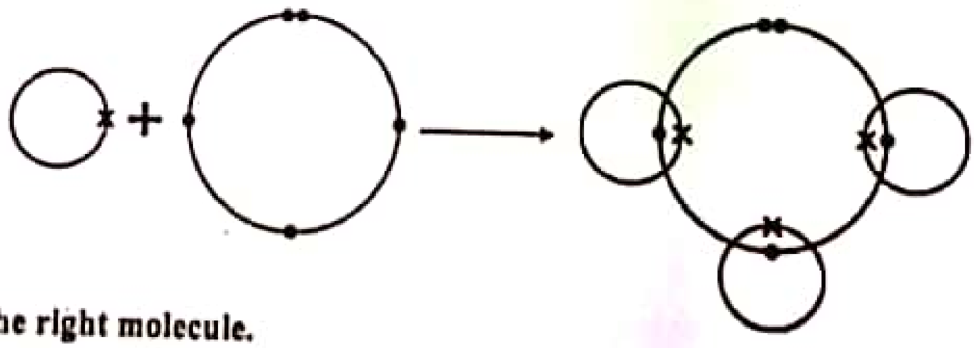
Q.15 Which one of the following hydrogen bonds is stronger than others?

A) $N^{\delta-}—H^{\delta+}$ .... $N^{\delta-}—H^{\delta+}$
B) $F^{\delta-}—H^{\delta+}$ .... $F^{\delta-}—H^{\delta+}$✓
C) $O^{\delta-}—H^{\delta+}$ .... $O^{\delta-}—H^{\delta+}$
D) $N^{\delta-}—H^{\delta+}$ .... $O^{\delta-}—H^{\delta+}$

Q.16 Which of the following is the correct dot and cross diagram of bonding betwe chlorine atoms?



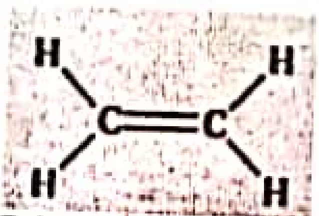
Q.17



Choose the right molecule.

A) CH3
B) CO
C) H2O
D) NH3✓

Q.18



Calculate the number of σ bonds and π bonds in the molecule.

A) 1π and 5σ bonds✓
B) 2π and 4σ bonds
C) 3π and 3σ bonds
D) 6 and 6σ bonds

Q.19 What is the exact value of angle in BF_3:

A) 90°
B) 119.5°
C) 104.5°
D) 120°✓

Q.20 Which option shows all the molecules with bond angle 109.5°:

A) CH_4, CCl_4, NH_3
B) CH_4, NH_4 PH_3
C) $SiCl_4$ H_2O $BeCl_2$
D) $SiCl_4$ NH_4CH_4✓

Chemical Energetics

Q.1 Which one of the following molecules has zero dipole moment?

A) NH_3
B) $CHCl_3$
C) BF_3✓
D) H_2O

Q.2 A spontaneous process is

A) Unidirectional and irreversible
B) Irreversible and a real process
C) Unidirectional and a real process
D) All of the above✓

Q.3 Which of the following is an exothermic reaction?

A) $H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(l)}$ ✓
B) $Na_{(g)} \rightarrow Na^+_{(g)} + 1e^-$
C) $\frac{1}{2}H_{2(g)} \rightarrow H_{(g)}$
D) $\frac{1}{2}Cl_{2(g)} \rightarrow Cl_{(g)}$

Q.4 Which of the following formation is an endothermic reaction?

A) $C_{(g)} + O_{2(g)} \longrightarrow CO_2$
B) $N_{2(g)} + 3H_{2(g)} \longrightarrow 2NH_{3(g)}$
C) $2H_2O_{(l)} \longrightarrow 2H_{2(g)} + O_{2(g)}$ ✓
D) None of the above

Q.5 The heat of hydration decreases with the increase in:

A) Number of neutrons
B) Size of cations✓
C) Size of atomic radii
D) Number of electrons

Q.6 Which of the following formation is endothermic reaction?

A) $2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$.
B) $C(s) + O_2(g) \longrightarrow CO_2(g)$.
C) $N_2(g) + O_2(g) \longrightarrow N_2O_2(g)$.✓
D) None of these.

Q.7 Which of the following bonds has minimum bond energy?

A) C – F.
B) C – Cl.
C) C – I.✓
D) C – Br.

Q.8 The amount of heat absorbed by one mole of solid at 1 atm when it melts into liquid form is denoted by ___________

A) Δ Hv.
B) Δ Hf.✓
C) Δ Hi.
D) Δ Hs.

Q.9 Lattice energy of an ionic crystal is the enthalpy of:

A) Combustion
B) Dissociation
C) Dissolution
D) Formation✓

Q.10 In standard enthalpy of atomization, heat of the surrounding:

A) Remains unchanged
B) Increases
C) Increases than decreases
D) Decreases✓

Solutions

Q.1 Which type of force is present in gasoline?
A) Dipole-dipole forces
B) Dipole-induced dipole forces
C) London dispersion forces✓
D) hydrogen bonding

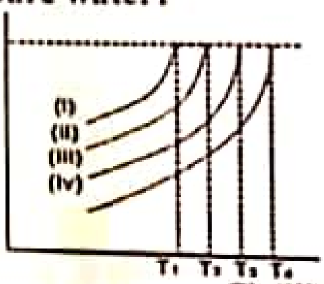
Q.2 The standard enthalpy of solution of NH4Cl is ____________ $kJmol^{-1}$.
A) +16.2✓
B) -25.0
C) +4.98
D) +26.0

Q.3 0.1 mole of acetic acid has been dissolved per dm^3 of the solution, the percen ionization of acetic acid will be
A) 13
B) 15
C) 1.3✓
D) 0.1

Q.4 Solubility of $Ce_2(SO_4)_3$
A) Increases with temperature
B) Decreases with temperature
C) Shows exceptional behavior✓
D) Remains constant

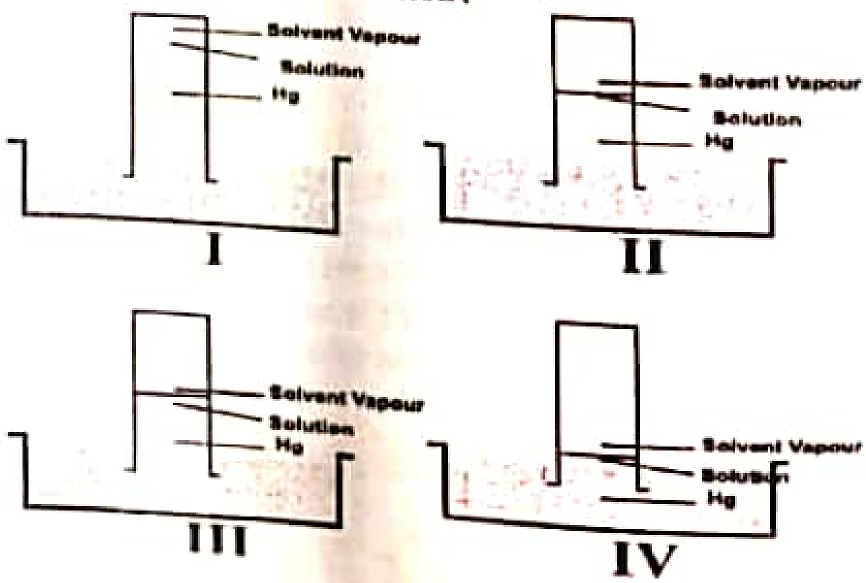
Q.5 Solubility of $KClO_3$ can be decreased bin H_2O by:
A) Removing K^+ ions from the solution
B) Removing $ClO3^{-1}$ ions from the solution
C) Adding KCl from outside✓
D) Adding NaNO3 from outside

Q.6 The vapor pressure lines for pure as well as solutions of different concentration shown. Which line represents pure water?



A) (i) ✓
B) (ii)
C) (iii)
D) (iv)

Q.7 One mole of glucose was dissolved in 1 kg of water, ethanol, ether and benzene sepa and the molal boiling point constant of each individual solution was found to be 1.75, 2.16 and 2.70 in the units of / °C kg mol^{-1} respectively. Which of the foll figures shows benzene as solvent in solution?



A) I✓
B) II
C) III
D) IV

Q.8 10.0 grams of glucose are dissolved in water to make 100 cm^3 of its solution, its molarity is:
A) 0.55✓
B) 0.1
C) 10
D) 1

Q.9 Given solution contains 16.0 g of CH_3OH, 92.0 g of C_2H_5OH and 36 g of water. Which statement about mole fraction of the components is true?
A) Mole fraction of CH_3OH is highest among all
B) Mole fraction of C_2H_5OH and H_2O is the same✓
C) Mole fraction of CH_3OH and C_2H_5OH is same components
D) Mole fraction of H2O is the lowest among all

Q.10 Freezing point will also be defined as that temperature at which its solid and liquid phases have the same:
A) Concentration
B) Ratio between the particles
C) Vapour pressure✓
D) Attraction between the phases

Electrochemistry

Q.1 Which of the following compounds, in the form of aqueous solution, on reaction with sodium carbonate will produce carbon dioxide gas?
A) H_3C-COO-C_2H_5
B) H_3C_2-COO-CH_3
C) H3C_2-CO-OH✓
D) H_3C_2-COO-C_2H_5

Q.2 Plasma is the ionized gas mixture which consists of
A) Ions and electrons
B) Electrons and neutral atoms
C) Electrons, ions and neutral atoms✓
D) Ions and neutral atoms

Q.3 In conjugated protein molecules, the protein is attached or conjugated to some non-protein group which are called:
A) Prosthetic Group✓
B) Aldehyde Group
C) Hydrogen Bonding
D) Peptide Linkage

Q.4 Stronger the oxidizing agent, greater is the:
A) Redox Potential
B) emf of the cell
C) Oxidation Potential
D) Reduction Potential✓

Q.5 The emf produced by Galvanic Cell is known as:
A) Redox Potential
B) Oxidation Potential
C) Cell Potential✓
D) None of the above

Q.6 In nickel-cadmium battery, the cathode is composed of:
A) Cd
B) $Ni(OH)_2$
C) Ni
D) NiO_2✓

Q.7 Concentrated sugar solution undergoes hydrolysis into glucose and fructose by enzyme called:
A) Zymase
B) Invertase✓
C) Cellulose
D) Urease

Q.8 The most durable metal plating on iron to protect against corrosion is:
A) Tin plating
B) Zinc plating✓
C) Nickel plating
D) Copper plating

B) Bromoform
D) Chloroform✓

Q.7 Peroxyacetyl nitrate is an irritant to human beings and it effects:
A) Nose
B) Stomach
C) Ears
D) Eyes✓

Q.8 The increase in concentration of oxidizing agents in smog like H_2O_2 HNO_3, PAN and ozone in the air is called
A) Carbonated smog
B) Nitrated smog
C) Photochemical smog✓
D) Sulphonated smog

Q.9 Which is the metal, whose elevated concentration is harmful for fish as it clogs the gills thus causing suffocation?
A) Sodium
B) Lead
C) Zinc
D) Aluminium✓

Q.10 Aerobic decomposition of organic matter i.e. glucose by bacteria in water sediments produces:
A) Propene
B) Ethane
C) Methane✓
D) Butane

Equilibrium constant Kc for

$$H_2O \longrightarrow H^+ + OH^-$$

Can be written as follows:

A) $K_c = \frac{[H^+]}{[H_2][OH^-]}$

B) $K_c = \frac{[OH^-]}{[H^+][OH^-]}$

C) $K_c = \frac{[OH^-][H^+]}{[H_2O]}$ ✓

D) $K_c = \frac{[H_2O]}{[H^+][OH^-]}$

________ is used as catalyst in Haber's process for NH3 gas manufacture.

A) Iron.✓

B) Carbon.

C) Copper.

D) Silver.

In 'AgCl' solution. Some salt of NaCl is added, 'AgCl' will be precipitated due to:

A) Solubility

B) Electrolyte

C) Unsaturation effect

D) Common ion effect✓

'Ka' for an acid is higher, the stronger is the acid; relate the strength an acid with 'pKa'

A) Higher pKa, weaker the acid

C) pKa has no relation with acid strength

B) Lower pKa, stronger the acid

D) Both A and B✓

Formation of NH_3 is reversible and exothermic process, what will happen on cooling?

A) More reactant will form

B) More N_2 will be formed

C) More H_2 will be formed

D) More product (NH_3) will be formed✓

A buffer solution is that which resists/minimizes the change in

A) pOH

B) pH✓

C) pKa

D) pKb

The chemical substance, when dissolved in water, gives "H^+" is called:

A) Acid✓

B) Base

C) Amphoteric

D) Neutral

The 'pH' of our blood is:

A) 6.7 – 8

B) 7.9

C) 7.5

D) 7.35 – 7.4✓

The value of equilibrium constant (Kc) for the reaction $2HF_{(g)} \rightleftharpoons H_{2(g)} + F2(g)$ is 10^{-13} at 2000 °C. Calculate the value of Kp for this reaction:

A) 2×10^{-13}

B) 10^{-13}✓

C) 186×10^{-13}

D) 3.48×10^{-9}

What will be the pH of a solution of NaOH with a concentration of 10^{-3} M?

A) 3

B) 14

C) 11✓

D) 7

Reaction Kinetics

Q.1 The reaction which is responsible for the production of electricity in the Voltaic cell is:
A) Hydrolysis reaction
B) Oxidation reaction
C) Redox reaction✓
D) Reduction reaction

Q.2 The rate of reaction involving ions can be studied by ________ method
A) Dilatometric
B) Refractometric
C) Optical rotation
D) Electrical conductivity✓

Q.3 The Kc has following units for the reaction $H_2(g) + I_{2(g)} \rightleftharpoons 2HI_{(g)}$
A) mol^3dm_{-3}
3B) $moldm^3$
C) mol^1dm
D) No unit✓

Q.4 The rate equation determined experimentally for this reaction:
(CH3)3—C—Br + H2O (CH3)3—C—OH + HBr
Is, Rate = k[(CH3)3CBr]
Hence it is which of the follwing?
A) Fractional Order
B) Pseudo First Order✓
C) First Order
D) Second Order

Q.5 A limiting reactant is the one which:
A) Is mostly a cheaper substance and taken in larger quantity
B) Is consumed earlier and controls the amount of product formed in a chemic reaction✓
C) Gives greatest number of moles of products
D) Is left behind after the completion of reaction

Q.6 ________ is colorless volatile liquid at room temperature.
A) HCl. B) HI. C) HF.✓ D) HBr.

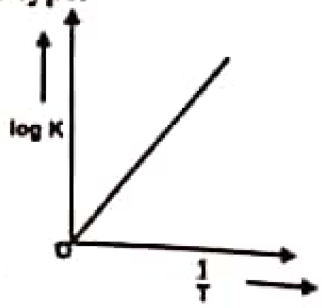
Q.7 It is experimentally found that a catalyst is used to:
A) Lower the activation energy✓
B) Increase the activation energy
C) Lower the pH
D) Decrease the temp of the reaction

Q.8 According to collision theory of bimolecular reaction sin gas phase, the minimum amo of energy required for an effective collision is known as:
A) Heat of reaction
B) Rate of reaction
C) Has no effect on the reaction
D) Energy of activation✓

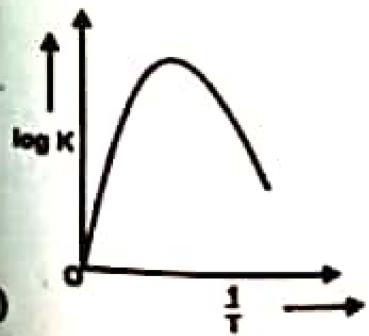
Q.9 In some reactions, a product formed acts as a catalyst. The phenomenon is called
A) Negative Catalysis
B) Activation of Catalyst
C) Hetergeneous catalysis
D) Autocatalysis✓

Q.10 The reaction rate in forward direction decreases with the passage of time because
A) Concentration of reactants decrease✓
B) Concentration of product decreases
C) The order of reaction changes
D) Temperature of the system changes

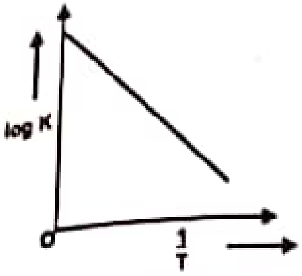
Q.11 By considering Arrhenius equation, the graph between '$\frac{1}{T}$' and 'log K' given a curve the type:

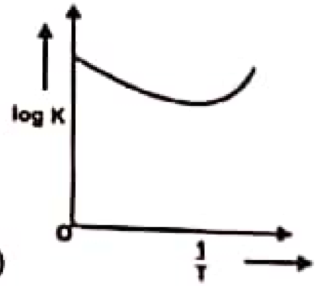


A)



B)

C)


D)


Q.12 **In zero order reactions, the rate is independent of:**
A) Concentration of the product
B) Concentration of the reactant✓
C) Temperature of the reaction
D) Surface area of the product

Q.13 **If the reactant or product of a chemical reaction can absorb ultraviolet, visible or infrared radiation, then the rate of a chemical reaction can best be measured by which one of the following methods?**
A) Chemical method
B) Spectrometry✓
C) Graphical method
D) Differential method

Q.14 **For the reaction $2NO + O_2 \rightleftharpoons 2NO_2$, the rate equation for the forward reaction is**
A) Rate = k $[NO][O_2]$
B) Rate = k $[NO]^2 [O_2]$ ✓
C) Rate = k $[NO_2]^2$
D) Rate = k $[NO_2]$

Q.15 **In the half-life of an element, the equation for the number of decaying atoms is given by**
A) $\Delta \propto - N\Delta t$✓
B) $\Delta N = KN\Delta t$
C) $\Delta N \propto -n\Delta t$
D) $\Delta N = -\Delta N\Delta t$

Q.16 **When the change in concentration is 6×10^{-4} mol dm^{-3} and time for that change is 10 seconds, the rate of reaction will be**
A) 6×10^{-3} mol dm^{-3} sec^{-1}
B) 6×10^{-4} mol dm^{-3} se^{-1}
C) 6×10^{-2} mol dm^{-3} sec^{-1}
D) 6×10^{-5} mol dm^{-3} sec^{-1}✓

If the reactant 'B' is in excess, the order of reaction with respect to 'A' in given rate law, Rate = $k[A]^2[B]$ is:
A) 2nd order reaction✓
B) 1st order reaction
C) Pseudo 1st order reaction
D) 3rd order reaction

Q.17 **The rate constant 'k' is 0.693 min^{-1}. The half-life for the 1st order reaction will be:**
A) 1 min✓ B) 0.693 min C) 2 min D) 4 min

Q.18 **Rate of first order reaction depends on__________:**
A) Concentration of one reactant✓
B) Concentration of two reactants
C) Concentration of three reactants
D) Independent of the initial concentration

Q.19 **Gas is enclosed in a container of 20cm^3 with the moving piston. According to kinetic theory of gases what will be the effect on freely moving molecules of the gas if temperature is increased from 20°C in 100°C?:**
A) Volume will be increased✓
B) Pressure will become one half
C) Temprature has no effect on freely moving molecules
D) Colliding capability of molecule will become lower

Q.20 **Role of a catalyst in a chemical reaction is to:**
A) increase rate of a reaction✓
B) decrease rate of a reaction
C) decreases yield of a reaction
D) increase yield of product

INORGANIC CHEMISTRY

Inorganic chemistry deals with synthesis and behavior of *inorganic* and organometallic compounds. This field covers all *chemical* compounds except the myriad of organic compounds (carbon-based compounds, usually containing C-H bonds), which are the subjects of organic *chemistry*.

Periods

Q.1 Which of the following sulphates is not soluble in water?
A) Sodium Sulphate
B) Barium Sulphate✓
C) Potassium Sulphate
D) Zinc Sulphate

Q.2 White lead has one of the following properties
A) Acidic
B) Crystalline
C) Amorphous✓
D) Neutral

Q.3 In Modern Periodic Table, the elements in Group II-B are:
A) Zn, Cd, Pb
B) Zn, Cd, Hg✓
C) Zn, Cd, Ba
D) Zn, Cd, Bi

Q.4 Hydrogen loses an electron to form:
A) H^+✓
B) $H2^{-2}$
C) H
D) H^-

Q.5 At which condition are hydrides of alkaline earth metals formed:
A) At high pressure✓
B) At room temperature
C) At high temperature
D) None of the above

Q.6 Which metal carbide is formed readily by the direct reaction?
A) Rubidium
B) Potassium
C) Sodium
D) Lithium✓

Q.7 Asbestos is hydrated ________ magnesium silicate.
A) Calcium✓
B) Aluminium
C) Barium
D) Carbon

Q.8 Which Noble Gas is used in bacterial lamps?
A) Xenon✓
B) Radon
C) Argon
D) Krypton

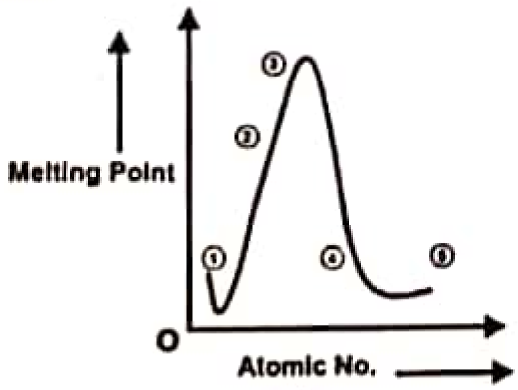
Q.9 The elements of IIIA to VIIIA subgroups except He are known as ________ block elements.
A) q.
B) s.
C) p.✓
D) None of these.

Q.10 Na may be denoted by ________ electron configuration notation
A) $1s^2\,2s^1$.
B) [Ar] $4s^1$.
C) [Ne] $3s^1$.✓
D) None of these.

Q.11 Which is the best drying agent in desiccators?
A) KOH.
B) Gypsum.
C) $CaCl_2$.✓
D) Silica sand.

Q.12 Carbon exists as allotropes, which are different crystalline or molecular forms of the same substance. Graphite and diamond are allotropes of carbon. Diamond is a non-conductor whereas graphite is a good conductor because:
A) Graphite has a layered structure
B) In graphite, all valence electrons are tetrahedrally bound
C) In graphite one of valence electron is free to move✓
D) Graphite is soft and greasy

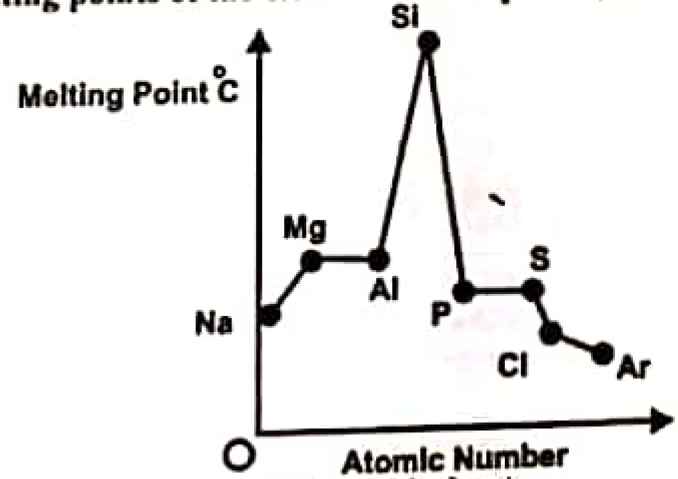
Q.13 The diagram below is a plot of melting points of elements of second period against their atomic numbers. Lithium and fluorine are placed at the extreme ends of the plot, on the basis of melting points where will you place Carbon among the empty slots on the plot?



A) 1
B) 2
C) 4
D) 3✓

Q.14 Which one remains same along a period?
A) Atomic radius
B) Melting point
C) Number of shells (orbits)✓
D) Electrical conductivity

Q.15 The trends, in melting points of the elements of 3rd period, are depicted in figure below.



The sharp decrease observed from 'Si' to 'P' is due to
A) Decrease in atomic radius from 'Si' to 'P'
B) Change in bonding and structure of two elements✓
C) Different universities of two elements
D) Increase in electron density from 'Si' to 'P'

Groups

Q.1 Which of the following carbonates of alkali metals is not stable towards heat and is decomposed on heating to its oxide along with liberation of CO2?
A) Li_2CO_3✓
B) Mg_2CO_3
C) K_2CO_3
D) Na_2CO_3

Q.2 The trend in the densities of elements of Group III-A of the Periodic Table is
A) A gradual increase✓
B) A gradual decrease
C) First decrease then increase
D) First increase then decrease

Q.3 $KMnO_4$ acts as a
A) Reducing agent
B) Excellent precipitating reagent
C) Germicide
D) Oxidizing agent✓

Q.4 In Modern Periodic Table, the elements in Group II-B are:
A) Zn, Cd, Pb
B) Zn, Cd, Hg✓
C) Zn, Cd, Ba
D) Zn, Cd, Bi

Q.5 At which condition are hydrides of alkaline earth metals formed:
A) At high pressure✓
B) At room temperature
C) At high temperature
D) None of the above

Q.6 Which metal carbide is formed readily by the direct reaction?
A) Rubidium
B) Potassium
C) Sodium
D) Lithium✓

Q.7 Asbestos is hydrated ________ magnesium silicate.
A) Calcium✓
B) Aluminium
C) Barium
D) Carbon

Q.8 In many of its properties ________ is quite different from the other alkali metals.
A) Li.✓
B) Be.
C) Na.
D) K.

Q.9 Name the rare halogen among the following.
A) F.
B) Cl.
C) I.
D) At.✓

Q.10 Concentrated nitric acid gives ________ when it reacts with tin.
A) Nitric oxide.
B) Meta stannic acid.
C) Ammonium nitrite.✓
D) None of these.

Elements of Biological Importance

Q.1 The presence of calcium is essential for the normal development of plants. An adequate supply of calcium appears to stimulate the development of which part of the plants?
A) Leaves
B) Fruits
C) Root hairs✓
D) Branches

Q.2 The noble gas which is used in radiotherapy of cancer is
A) Radon✓
B) Xenon
C) Krypton
D) Argon

Q.3 The coagulant used in raw water to precipitate suspended impurities is
A) Caustic soda
B) Lime water
C) Alum✓
D) Soda ash

Q.4 The whiteness of the recycled newspaper is improved by treating it with:
A) Sodium hydroxide
C) Super oxides

ORGANIC CHEMISTRY

Organic chemistry is the study of the structure, properties, composition, reactions, and preparation of carbon-containing compounds, which include not only hydrocarbons but also compounds with any number of other elements, including hydrogen (most compounds contain at least one carbon–hydrogen bond), nitrogen, oxygen, halogens, phosphorus, silicon, and sulfur. This branch of chemistry was originally limited to compounds produced by living organisms but has been broadened to include human-made substances such as plastics. The range of application of organic compounds is enormous and also includes, but is not limited to, pharmaceuticals, petrochemicals, food, explosives, paints, and cosmetics.

Fundamental Principles

Q.1 A gasoline of higher octane number can be obtained by

A) Oxidative cleavage
B) Thermal cracking
C) Catalytic cracking✓
D) Steam cracking

Q.2 Ethyne molecule is formed when two carbon atoms joined together to form a sigma bond by

A) sp-s overlap
B) sp^3-sp^3 overlap
C) $2p_y$-2py overlap
D) sp-sp overlap✓

Q.3 The formula of acrylonitrile is:

A) CH_2=CH—CN✓
B) CH_3—CH_2—CH_2—CN
C) CH_3—CH_2—CN
D) CH_3—CN

Q.4 The compound with an atom, which has unshared pair of electrons is called:

A) Nucleophile✓
B) Electrophile
C) Protophile
D) None of the above

Q.5 1-chloropropane and 2-chlorpropane are isomers of each other, the type of isomerism in these two is called:

A) Cis-trans isomerism
B) Chain isomerism
C) Position isomerism✓
D) Functional group isomerism

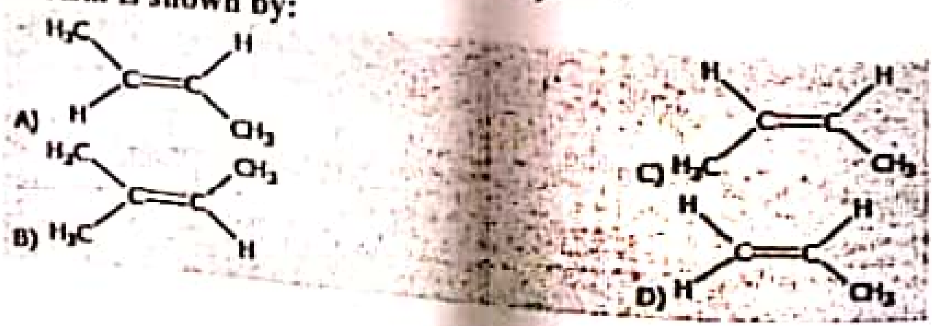
Q.6 Ethene on polymerization, gives the product polyethene. This reaction may be called as

A) Addition✓
B) Condensation
C) Substitution
D) Pyrolysis

Q.7 In the following, which one is free radical?

A) Cl^-
B) Cl'
C) Cl2
D) $Cl^•$✓

Q.8 The cis-isomerism is shown by:



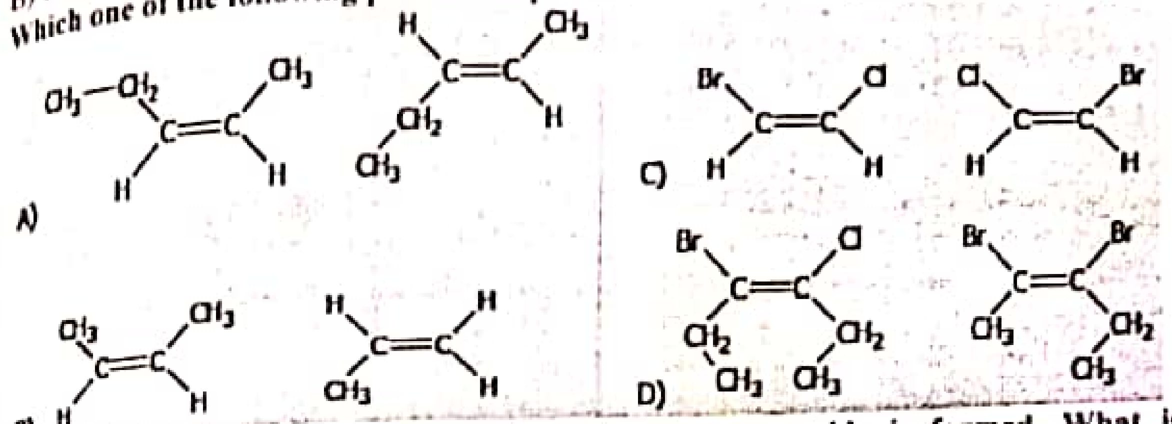
Q.9 Select the nucleophile from the following examples:

A) NO_2
B) NH_3✓
C) NO_2^+
D) N^+H_4

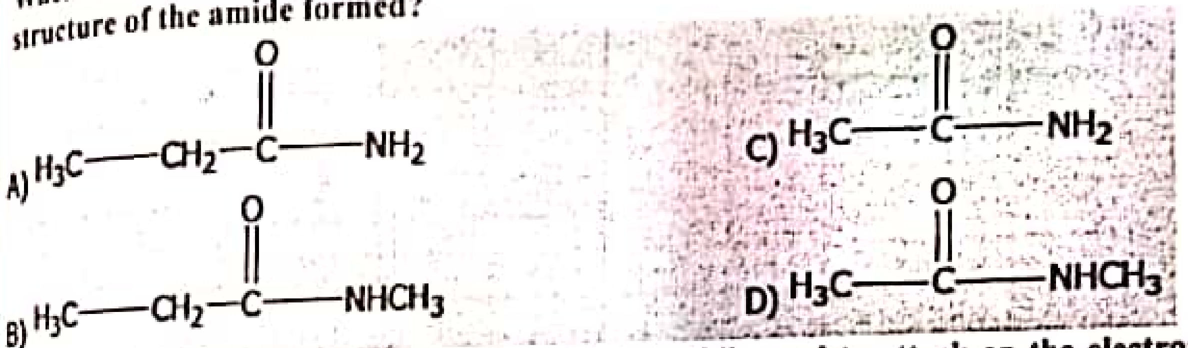
Q.10 Which one of the following compound is a ketone?

A) $CH_3-O-CH_2-CH_3$✓
B) $CH_3-CO-CH_2-CH_3$
C) $CH_3COCOOH$
D) CH_3-CH_2CHO

Q.11 Which one of the following pair of compounds is cis and trans isomers of each other?



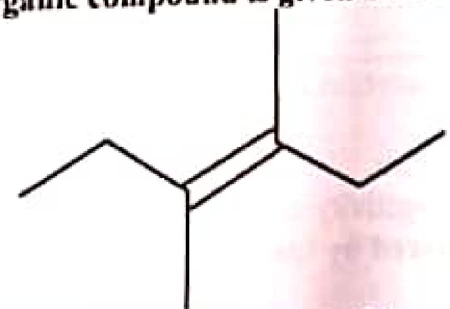
Q.12 When ethanol chloride reacts with methylamine, an amide is formed. What is the structure of the amide formed?



Q.13 Which one of the following is a powerful electrophile used to attack on the electrons of benzene ring?

A) $FeCl_3$
B) $FeCl_4^-$
C) Cl^+✓
D) Cl_2

Q.14 Skeletal formula of an organic compound is given below:



It is a hydrocarbon. IUPAC name of the compound is:

A) 3, 3-dimethyl-3-hexene
B) 3, 4-dimethyl-3-hexene✓
C) 3-hexene
D) 2,3-dimethyl-1-hexene

Q.15 Which one of the following pairs can be cis-trans isomer to each other?

A) $CHCl=CCl_2$ and $CH_2=CH_2$
B) $CHCl=CH_2$ and $CH_2=CHCl$
C) $CH_3-CH=CH-CH_3$ and $H_3CCH=CH-CH_3$✓
D) CH_3-CH_3 and $CH_2=CH_2$

Hydrocarbon

Q.1 The introduction of NO2 group in benzene ring is called 'Nitration'. The nitration of benzene takes place when it is heated with a 1:1 mixture of _______ at 50 °C-55 °C.
A) Conc. HNO_3 and conc. HCl
B) Conc. HNO_3 and conc. Acetic acid
C) Conc. HNO_3 and H_3PO_4
D) Conc. HNO_3 and conc. H_2SO_4✓

Q.2 Sublimation is used to purify
A) Ammonium sulphate
B) Sodium chloride
C) Benzoic acid✓
D) Lead carbonate

Q.3 During nitration of benzene the active nitrating agent is:
A) NO_2^- B) NO3 C) HNO2 D) NO_2^+✓

Q.4 Which compound is the most reactive one?
A) Ethyne B) Benzene C) Ethane D) Ethene✓

Q.5 Which one of the following compounds show cis-trans isomerism?
A) 1-butene
B) 1-hexene
C) 1-bromo-2-chloropropane✓
D) Propene

Q.6 Which element forms long chains alternating with oxygen?
A) Carbon. B) Nitrogen. C) Silicon. D) All of these.✓

Q.7 Which bond will break when electrophile attacks an alcohol?
A) O – H.✓ B) Both A and B. C) C – O. D) None of these.

Q.8 The extent of un-saturation in a fat is expressed as its
A) Acid number.
B) Iodine number.✓
C) Saponification number.
D) None of these.

Q.9 Alkanes containing _______ carbon atoms are waxy solids.
A) up to 4.
B) 5 to 17.
C) 18 or more.✓
D) None of these.

Q.10 Hydrogen passed through phenol at 150 °C in the presence of _______ catalyst gi cyclohexanol.
A) Tin.
B) Nickel.✓
C) Iron.
D) Sodium.

Alkyl Halides

Q.1 During SN_2 reactions, configuration of the alkyl halide molecule:
A) Gets inverted✓
B) Remains same
C) Depends upon the carbon atom
D) Depends upon the electronegativity of halide

Q.2 Grignard reagents are prepared by the reaction of magnesium metal with alkyl halide the presence of
A) Dry Ether✓
B) Sodium Lead Alloy
C) Alcohol
D) Water

Q.3 Methanol is prepared from carbon monoxide and hydrogen. The catalyst used for reaction is
A) ZnO + CoO_2
B) ZnO + CuO
C) ZnO + Ag_2O
D) Cr_2O_3 + ZnO✓

Q.4 Ethanol reacts with Ammonia to produce ethyl amine, the catalyst is
A) $ZnCl_2$ C) C_6H_5N

B) ThO✓ D) Cr_2O_3

Q.5 Grignard reagents are prepared by the reaction of magnesium metal with alkyl halides in the presence of:

A) Dry Ether✓
B) CS_2
C) Alcohol
D) CCl_4

Q.6 When n-butyl magnesium iodide is treated with water, the product is:

A) n-butane✓
B) Iso-butane
C) Propane
D) Alcohol

Q.7 When purely alcoholic solution of sodium/potassium hydroxide and halogenoalkanes are reacted an alkene is formed, what is the mechanism of reaction?

A) Elimination✓
B) Dehydration
C) Debromination
D) Reduction of benzene

Q.8 The organic compound carbon tetrachloride is used as:

A) Lubricant
B) Solvent✓
C) Oxidant
D) Plastic

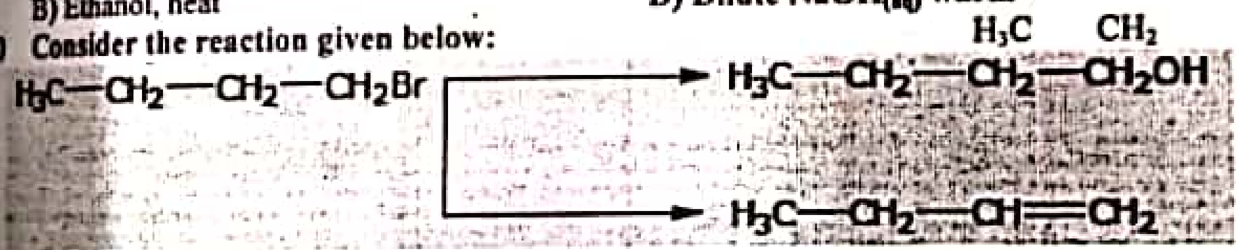
Q.9 The alkaline hydrolysis of bromoethane shown below gives alcohol as the product:

$$H_3C-CH_2-Br \longrightarrow H_3C-CH_2-OH$$

The reagent and the condition used in this reaction may be:

A) H_2O at room temperature
B) Ethanol, heat
C) KOH in alcohol
D) Dilute $NaOH_{(aq)}$ warm✓

Q.10 Consider the reaction given below:



Which statement is true?

A) Reagent for I is KOH in alcohol
B) Reagent for II is KOH in aqueous medium
C) Reaction I is Debromination
D) Reaction II is elimination✓

Alcohols and Phenols

Q.1 Dissociation constant of phenol is

A) 1.2×10^{-10}
B) 1.2×10^{10}
C) 1.3×10^{10}
D) 1.3×10^{-10}✓

Q.2 $CO + 2H2 \xrightarrow[Y]{X} CH_3OH$

X and Y are: Y

A) $ZnO + Al_2O_3$ and 450 °C: 200 atm
B) $ZnO + Cr_2O_3$ and 450 °C: 200 atm
C) Al2O3 + Cr_2O_3 and 200 °C: 200 atm
D) $ZnO + Cr_2O_3$ and 450 °C: 200 atm

Q.3 Phenol reacts with concentrated H2SO4 to give:

A) ortho hydroxy benzene sulphonic acid
B) meta hydroxy benzene sulphonic acid
C) ortho and para hydroxy benzene sulphonic acid✓
D) para hydroxy benzene sulphonic acid

Q.4 Phenol can be distinguished from alcohol by adding:

A) Br_2/H_2O✓
B) Cl_2/H_2O
C) $FeSO_4$
D) $FeCl_3$

Q.5 The products of the fermentation of a sugar are ethanol and
A) Water.
B) Oxygen.
C) Carbon dioxide.✓
D) Sulfur dioxide

Q.6 Name the partially miscible liquids from the following?
A) Alcohol-ether.
B) Nicotine-water.✓
C) Benzene-water.
D) Both A and B.

Q.7 Ethanol-water is ________ mixture.
A) Azeotropic.✓
B) Ideal.
C) Benedict's.
D) Aliphatic.

Q.8 CH_3-O-CH_3 is example of ________ isomerism.
A) Metamerism.
B) Functional group.✓
C) Chain.
D) Position.

Q.9 An alcohol is converted to an aldehyde with same number of carbon atoms as tha alcohol in the presence of $K_2Cr_2O_7/H_2SO_4$ the alcohol is:
A) $CH_3C(CH)_2OH$
B) $CH_3CH_2CH_2OH$✓
C) $(CH_3)_3COH$
D) $(CH_3)_2CHOH$

Q.10 Which enzyme is involved in the fermentation of glucose:
A) Zymase✓
B) Invertase
C) Urease
D) Diastase

Aldehydes and Ketones

Q.1 When a hydrogen atom is removed from benzene, the group left behind is called
A) Alkyl group
B) Phenyl group✓
C) Benzyl group
D) Methyl group

Q.2 Dry distillation of a mixture of calcium salts of formic acid and acetic acid results into th formation of
A) Formaldehyde
B) Acetaldehyde✓
C) Calcium acetate
D) Sodium acetate

Q.3 Brick red precipitates are formed when aldehydes react with
A) Sodium borohydride
B) Sodium bisulphite
C) Sodium nitroprusside
D) Fehling's solution✓

Q.4 In the conversion of ethylene into acetaldehyde, cupric chloride acts as:
A) Initiator
B) Promoter✓
C) Catalyst
D) Reactant

Q.5 When acetone is heated in the presence of $K_2Cr_2O_7/H_2SO_4$, the products formed are;
A) Maleic Acid and Fumaric Acid
B) Acetic Acid and Formic Acid✓
C) Formic Acid and Oxalic Acid
D) Oxalic Acid and Acetic Acid

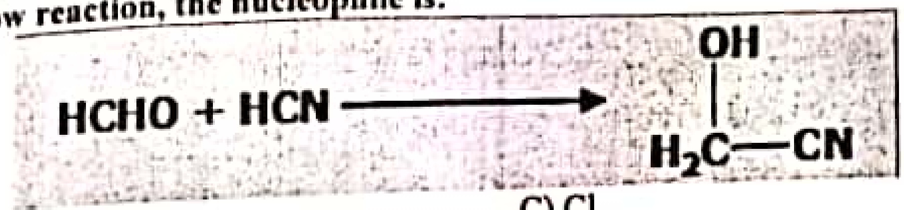
Q.6 Which of the following compounds will react with Tollen's Reagent?

Q.7 Aldehydes can be synthesized by the oxidation of
A) Primary alcohols.✓
B) Secondary alcohols.
C) Organic acids.
D) Inorganic acids.

Q.8 Which of the following is used to make chloral hydrate?
A) Acetaldehyde.✓
B) Formaldehyde.
C) None of these.
D) Both A and B.

Q.9 Consider the following reaction:
$R{-}CHO + {}_2[Ag(NH_3)_2]OH \longrightarrow R{-}COONH_4 + 2Ag + 2NH_3 + H_2O$
This reaction represents one of the following tests.
A) Fehling test
B) Benedict test
C) Ninhydrin test
D) Tollens test✓

Q.10 In the below reaction, the nucleophile is:



A) CN—✓
B) HCl
C) Cl
D) OH

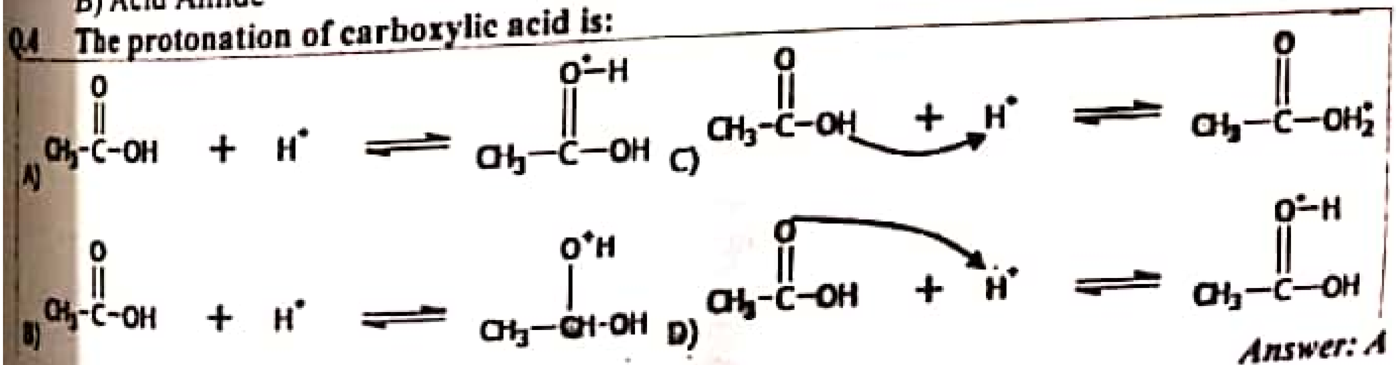
Carboxylic Acids

Q.1 The strongest acid among the following is
A) HF B) HCl C) HI✓ D) HBr

Q.2 Symmetrical alkanes can be produced by
A) Sabatier Sender's Reaction
B) Hydrogenolysis Reaction
C) Reduction Reaction
D) Kolbe's Electrolytic Reaction✓

Q.3 Hydrolysis of cyano group by an aqueous acid results into
A) Carboxylic Acid✓
B) Acid Amide
C) Cyanohydride
D) Formaldehyde

Q.4 The protonation of carboxylic acid is:



Answer: A

Q.5 Acetic acid is called __________ acid.
A) Methanoic. B) Ethanoic.✓ C) Propanoic. D) Butanoic.

Q.6 $CH_3COOH + PCl_5 \longrightarrow ?$
The products of the above reaction are:
A) $CH_3COCl + POCl_3 + HCl$✓
B) $CH_3COCl + POCl_2 + HCl$
C) $CH_3Cl + POCl_3 + HCl$
D) $CH_3COCl + POCl_3 + H_2$

Q.7 $CH_3CN + HCl \longrightarrow A + B$ (in the presence of water)

In the above reaction, A and B are:
A) Acetic acid and acid amide
B) Acetic acid and ammonia
C) Acetic acid and methyl chloride
D) Acetic acid and ammonium chloride✓

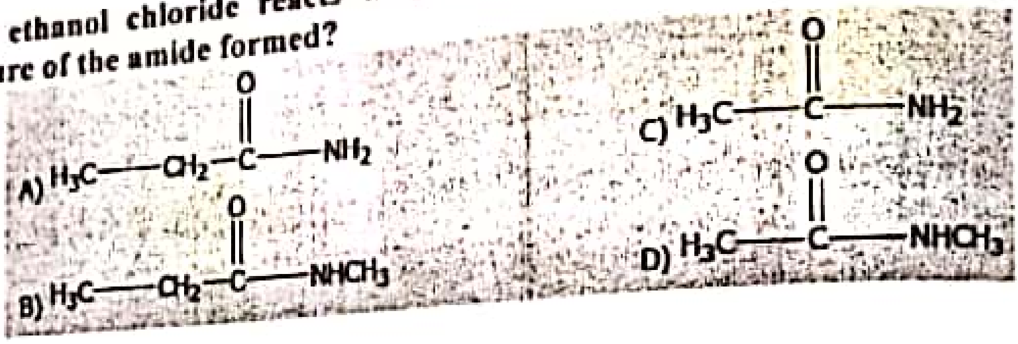
Q.8 Consider the following reaction:

$$CH_3COOH + Mg\ (metal) \longrightarrow ?$$

What product will form?
A) Magnesium formate
B) Magnesium acetate✓
C) Magnesium ion
D) Carboxylate ion

Q.9 In the below reaction the nucleophile which attacks on the carbon atom of acid is:
A) OH–
B) P
C) Cl–✓
D) H–

Q.10 When ethanol chloride reacts with methylamine, an amide is formed. What is structure of the amide formed?

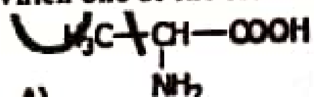


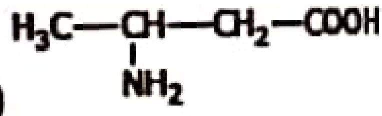
Amino Acids

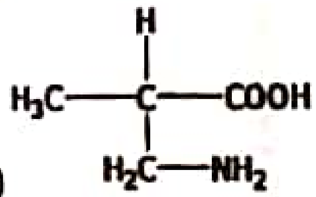
Q.1 The nature of the amino acid 'lysine' is
A) Neutral B) Amphoteric C) Acidic D) Basic✓

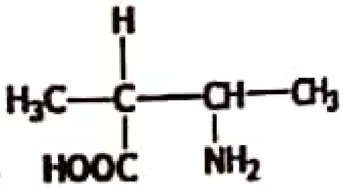
Q.2 The —NH—CO is called:
A) Amide group✓ B) Protein linkage C) Amino group D) Peptide linkage

Q.3 Which one of the following is an alpha amino acid?

A) 

C) 

B) 

D) 

Q.4 Which of the following has an amino R-group?
A) Lysine✓
B) Proline
C) Valine
D) Alanine

Q.5 At intermediate value of pH, amino acids form Zwitter ions containing:
A) $—N^+H_3$ and COO—✓
B) $—NH_3$ and COO—
C) $—N^+H_3$ and COOH
D) $—NH_3$ and COOH

Macromolecules

Q.1 Glucose is converted into ethanol by the enzyme ________ present in yeast:
A) Urease
C) Sucrase

B) Invertase D) Zymase✓

Q.2 The catalyst used for the preparation of acrylonitrile is

A) Cu_2Cl_2 and NH_4Cl✓ C) Cu_2Cl_2 and NH_4OH

B) Al_2O_3 and NH_4Cl D) Cu_2Cl_2 and Al_2O_3

Q.3 The calcium sulpho-aluminate is

A) $Co.Al_2O_3.3CaSO_4.6H_2O$ C) $3Ca.Al_2O_3.3CaSO_4.2H_2O$✓

B) $3Ca.Al_2O_3.CaSO_4.2H_2O$ D) $3Ca.Al_2O_3.3CaSO_4.6H_2O$

Q.4 Each molecule of haemoglobin is made up of nearly:

A) 11000 atoms C) 10000 atoms✓

B) 6600 atoms D) 6800 atoms

Q.5 Which acid is used in the manufacture of plastics?

A) Carbolic Acid C) Carbonic Acid✓

B) Acetic Acid D) Oxalic Acid

Q.6 In synthetic fibres ________ bonding is responsible for tensile strength.

A) Nitrogen. C) Oxygen.

B) Hydrogen. D) None of these.✓

Q.7 ________ are product of reaction of an alcohol and aromatic bi-functional acids.

A) Acrylic resins. C) PVCs.

B) Polyester resins.✓ D) Polyamide resins.

Q.8 When hexane dioic acid is heated with hexamethylene diamine, the compound formed is:

A) Polypeptide C) Ester

B) Addition polymer D) Nylon 6,6✓

Q.9 Glucose and fructose are common examples of:

A) Pentoses C) Heptoses

B) Hexoses✓ D) Butoses

Environmental Chemistry

Q.1 Seawater has 5.65×10^{-3} g of dissolved oxygen in one kilogram of water. Concentration of O2 in parts per million is

A) 5.65✓ C) 5.20

B) 7.69 D) 4.11

Q.2 Micronutrients are required in quantity ranging from:

A) 6 – 200 g per acre✓ B) 4 – 40 g per acre

C) 6 – 200 kg per acre D) 4 – 40 kg per acre

Q.3 Potassium fertilizers are especially useful for:

A) Mango C) Wheat

B) Tobacco✓ D) Rice

Q.4 The yellowish colour of photochemical smog is due to the presence of:

A) Nitrogen dioxide✓ C) Nitrous oxide

B) Dinitrogen trioxide D) Nitric oxide

Q.5 The incarnation process can reduce the volume of the water by:

A) One half C) One third

B) Not affected D) Two third✓

Q.6 The suspected liver carcinogen which also has negative reproduction and developmental effect on humans is:

A) Iodoform C) Tropoform